

A5 DPO 实验报告

作业：Assignment 5 - DPO for LLM Human Alignment

模型：GPT-2 Medium 355M SFT -> DPO

初始权重：/data/student/Fengjunming/Temp/26Spring-NLP-CS310/Lab/week11/gpt2-355M-sft.pth

训练数据：instruction-data-with-preference.json，共 1100 条偏好样本

训练配置：effective batch size = 8, AdamW lr = $1e-6$, weight decay = 0.01, beta = 0.1, 训练 320 step

DPO 目标：提升 policy 对 chosen response 的相对概率，同时用 reference model 约束偏移幅度。

最终训练 loss：0.6915

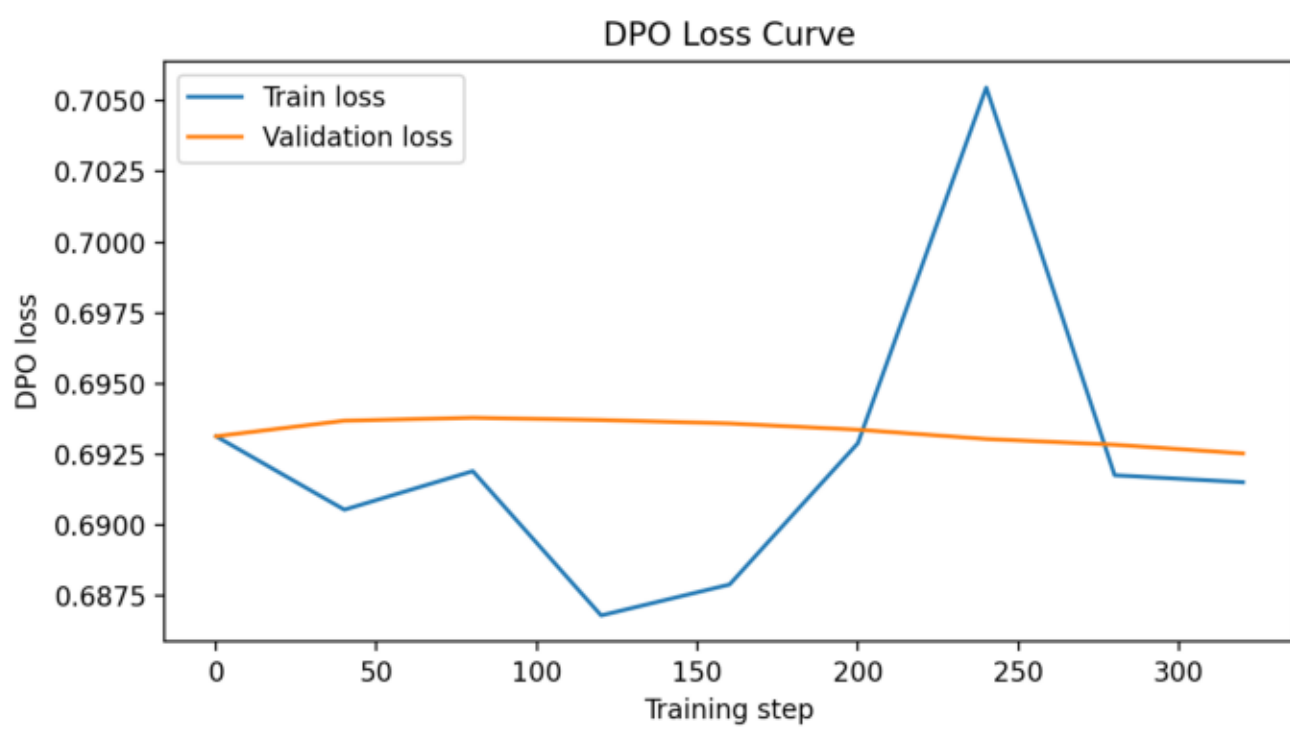
最终验证 loss：0.6925

最终训练 reward margin：0.0326

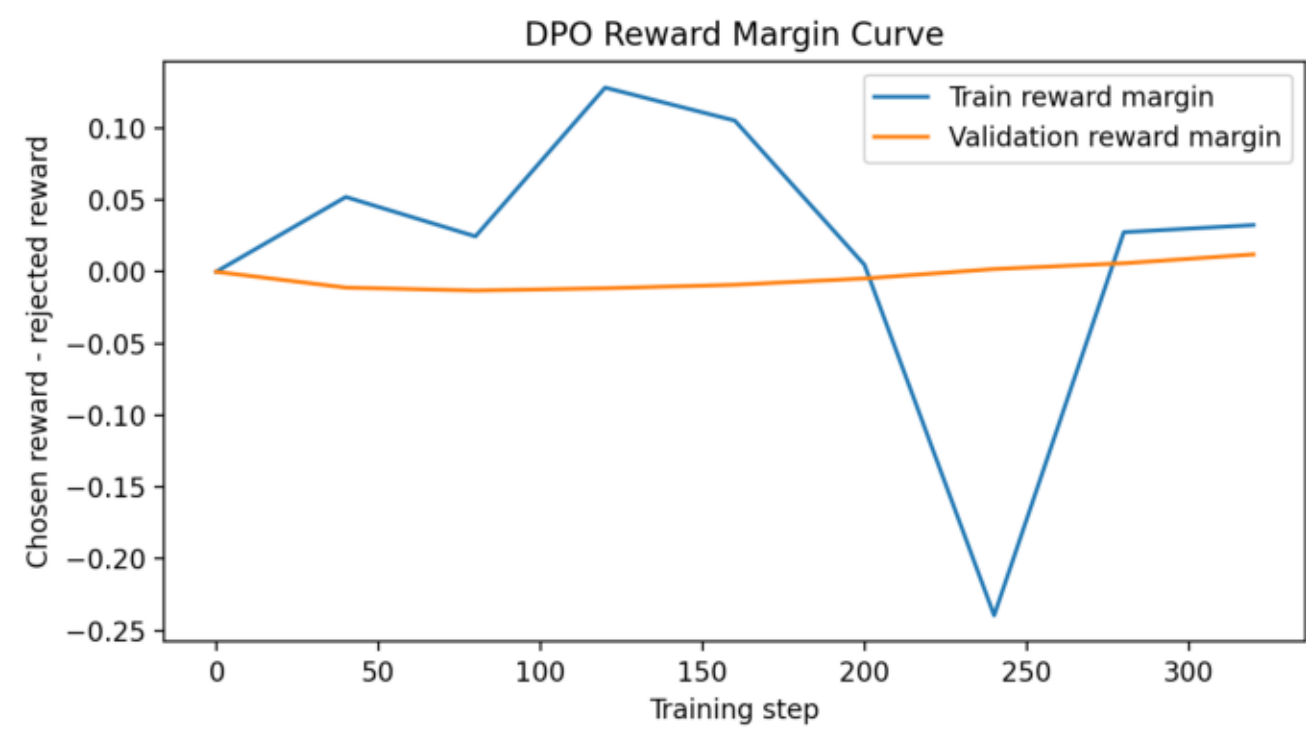
最终验证 reward margin：0.0121

保存文件：gpt2-medium355M-dpo.pth、dpo_tracking.json、dpo_loss_curve.png、dpo_reward_margin_cu

DPO Loss 曲线



DPO Reward Margin 曲线



AlpacaEval 结果截图

alpaca_eval evaluate result												
Unnamed: 0	win rate	standard error	mode	avg_length	n_wins	n_wins_base	n_draws	n_total	discrete_win_rate	length_controlled_winrate	lc_standard_error	
gpt2-medium355M-dpo.pth	62.611277	1.579213	community	76	501	296	8	885	62.732919	82.713607	0.09259	

评测结果

AlpacaEval/Qwen judge 结果

评测输入：model_outputs.json

参考输出：reference_outputs.json

评测器配置：qwen_judge

生成配置：without_response prompt, greedy decoding, max_new_tokens = 32

离线修复：AlpacaEval 默认 LC 指标使用本地 df_gamed.csv，避免 HuggingFace 下载失败。

leaderboard.csv 路径：/data/student/Fengjunming/Temp/26Spring-NLP-CS310/Assignment/A5/a5_dpo_co

	Unnamed: 0	win_rate	standard_error	mode	avg_length	n_wins	n_wins_base	n_draws	n_total	c
gpt2-medium355M-dpo.pth	62.611277	1.579213	community	76	501	296	8	805	62.732	